

Unit 7 & 8: Part 3

Efficiency and fairness



UNIT 7 & 8: Part 3 Overview

- Gains from trade
- Consumer and producer surplus
- Competitive markets lead to efficient allocations
- Deadweight loss – a measure of inefficiency if the quantity is different from the efficient quantity
- Price-setting firms and inefficient outcomes

Gains from trade

- When economic interactions are voluntary, they generally benefit both participants.
- The buyer will pay a price up to her willingness to pay (WTP).
 - WTP is determined by her income, prices of other goods and her preferences (i.e. utility gain from the item relative to alternative uses of the money).
- The seller will accept a price as low as his willingness to accept (WTA).
 - This is determined by the costs of providing the item, which in turn depends on technology and the prices of inputs.
- A buyer and seller of an item can both benefit if the value of the item to the buyer is more than the costs to the seller of supplying the item.
 - $WTP > WTA$

Gains from trade

- The difference $WTP - WTA$ is the **surplus** created by the transaction and it measures the **gains from trade**.
- How the surplus is divided between the seller and the buyer depends on the **price (P)**:
 - Consumer surplus = $WTP - P$
 - Producer surplus = $P - WTA$
- Surpluses can be regarded as rents, and they are an incentive to trade
 - participants are better off compared to the reservation option (not trading)
- When surplus is positive,
 - rational participants will choose to trade if a mutually beneficial price can be agreed upon.
 - trade would be a Pareto improvement.
 - failing to exploit gains from trade would be **inefficient**.



What about fairness?

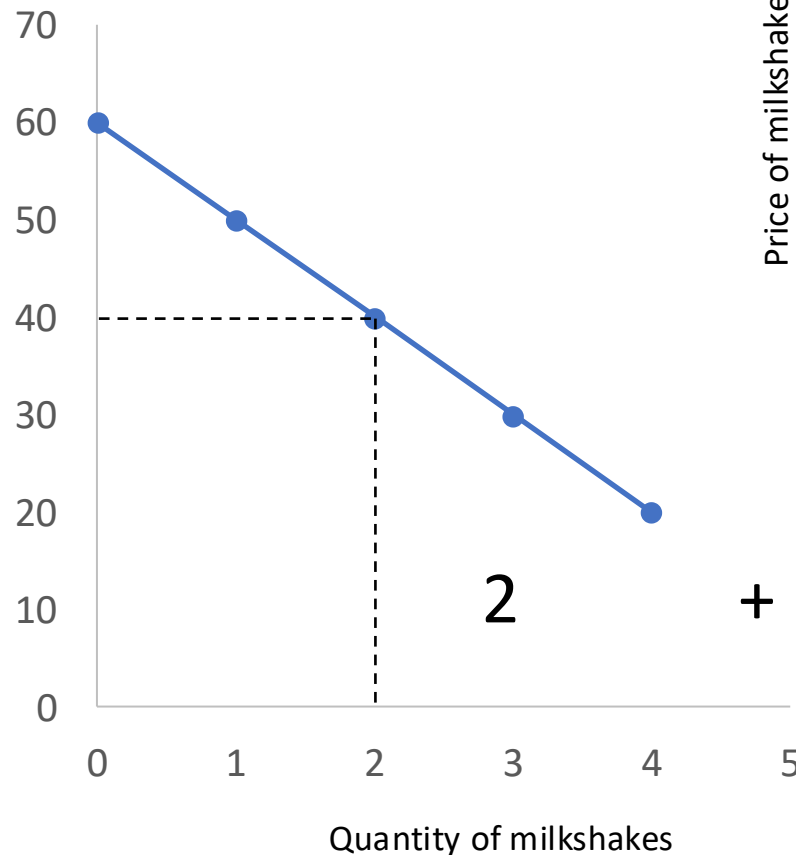
Two issues arise:

- The gains to the buyer and the seller may be distributed unfairly
- Not everyone can have the same opportunity to benefit from trade in markets
 - A person with no money/ resources cannot get gains from trade!

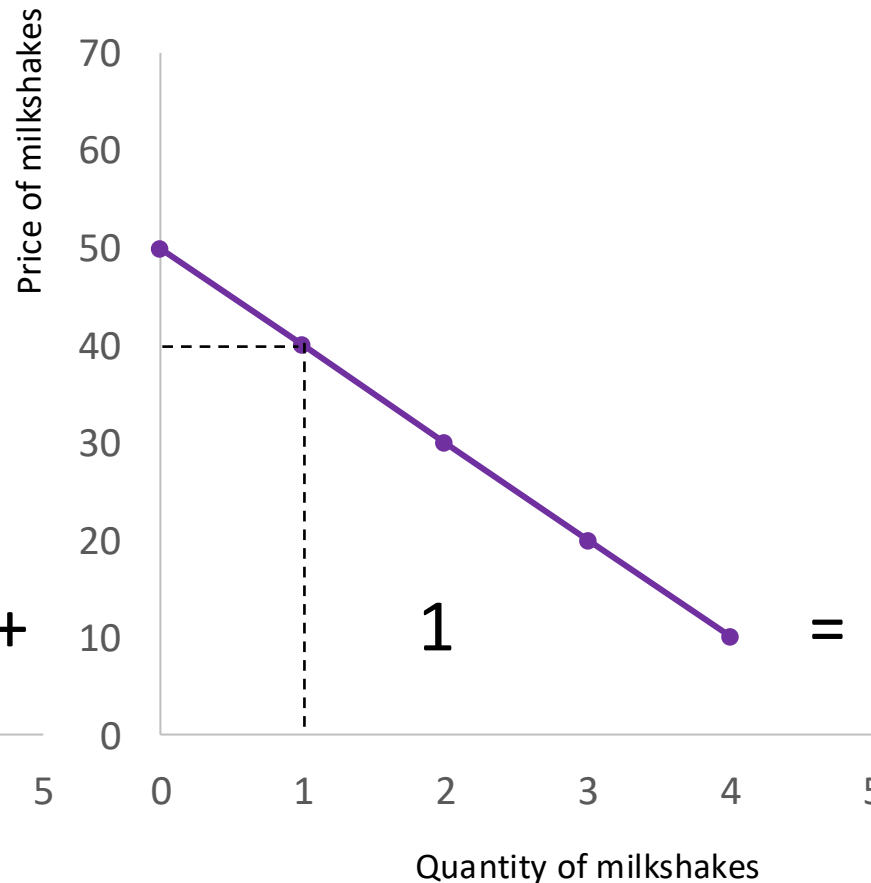
Gains from trade

- In markets there can be many buyers and sellers
- Buyers may differ in their WTPs for different units

Demand curve of consumer A



Demand curve of consumer B



Total demand

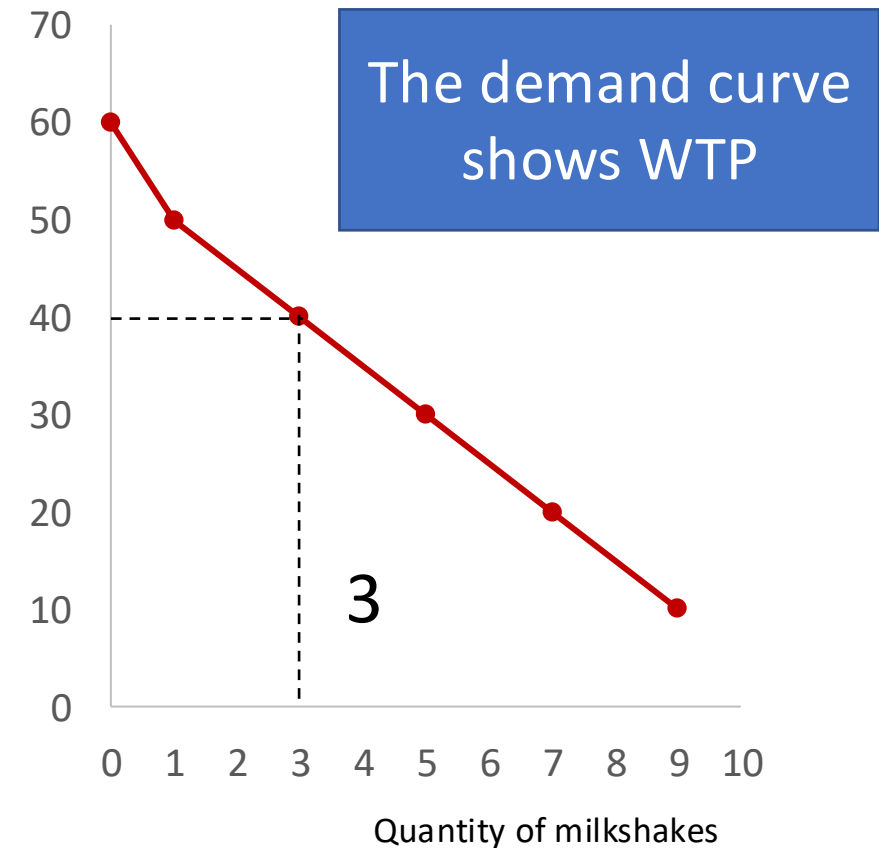
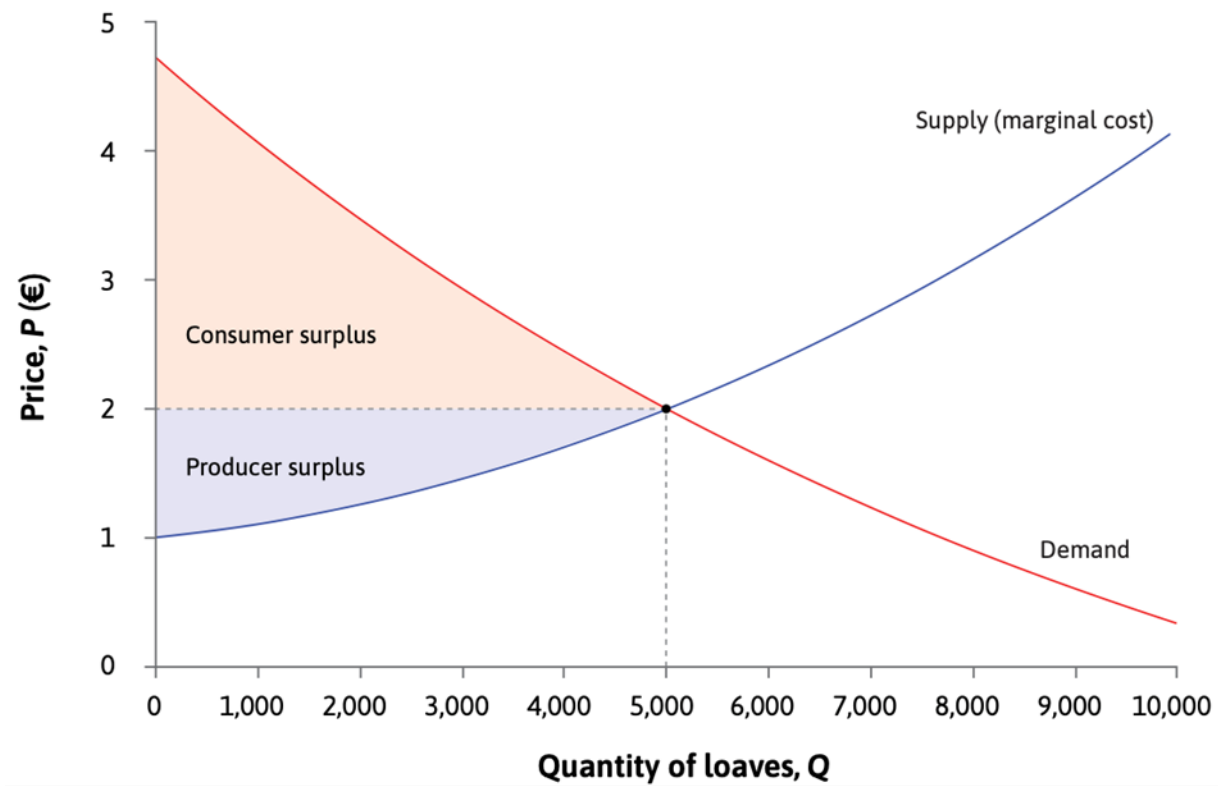
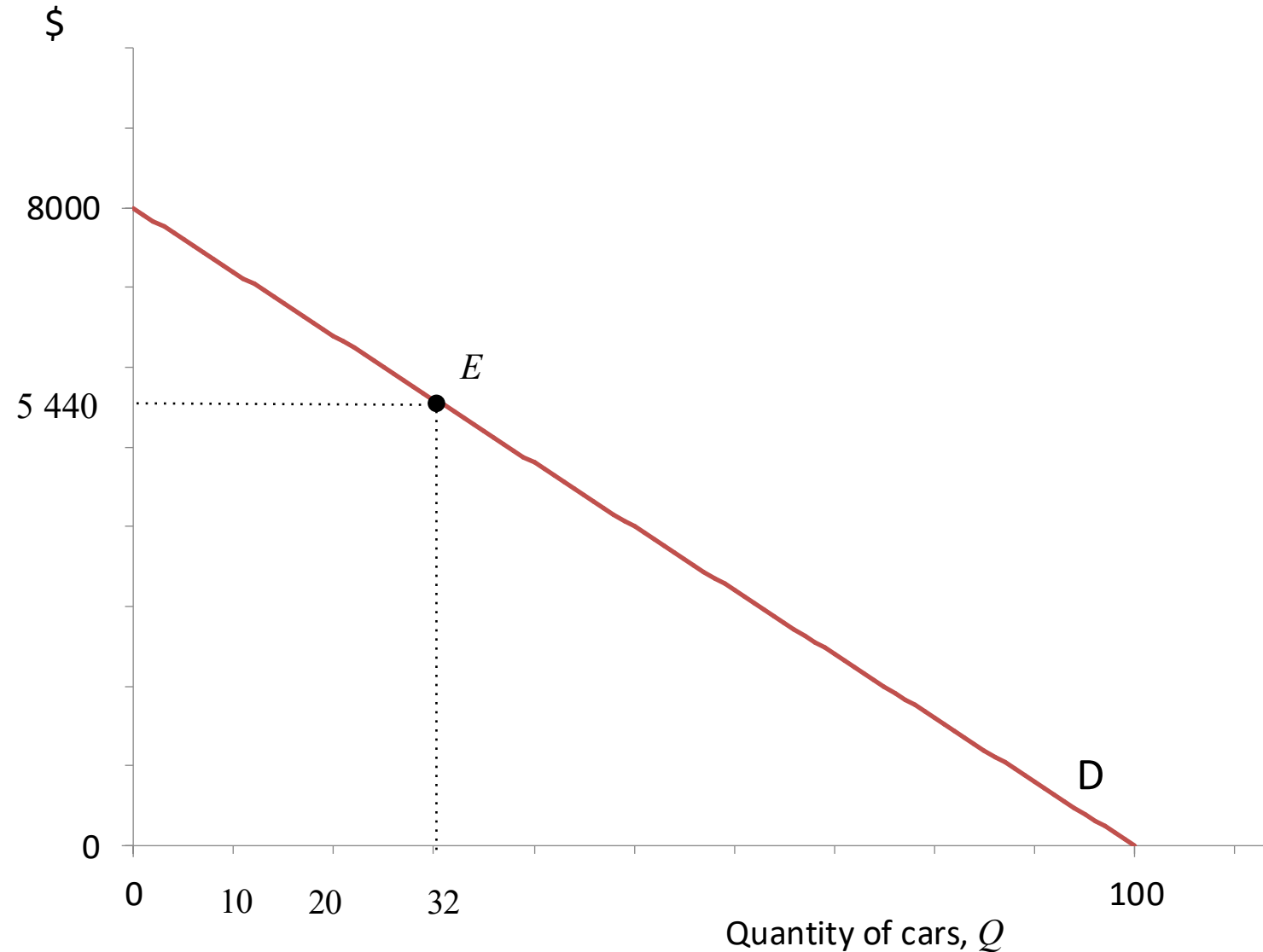


Fig 8.12



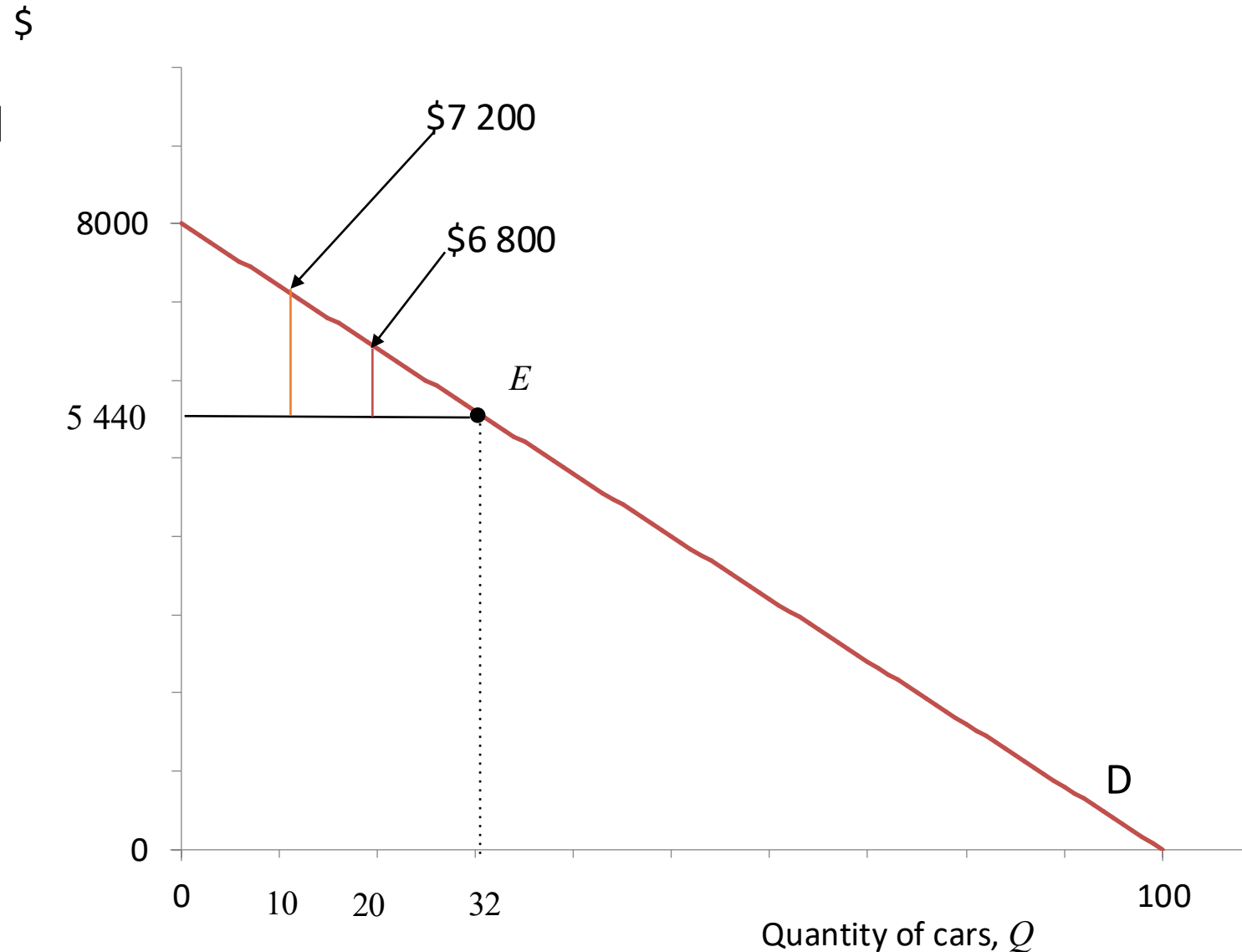
Consumer surplus

- Demand curve (D) is willingness to pay (WTP)
- Suppose 32 cars are sold at price of \$5 440 each



Consumer surplus

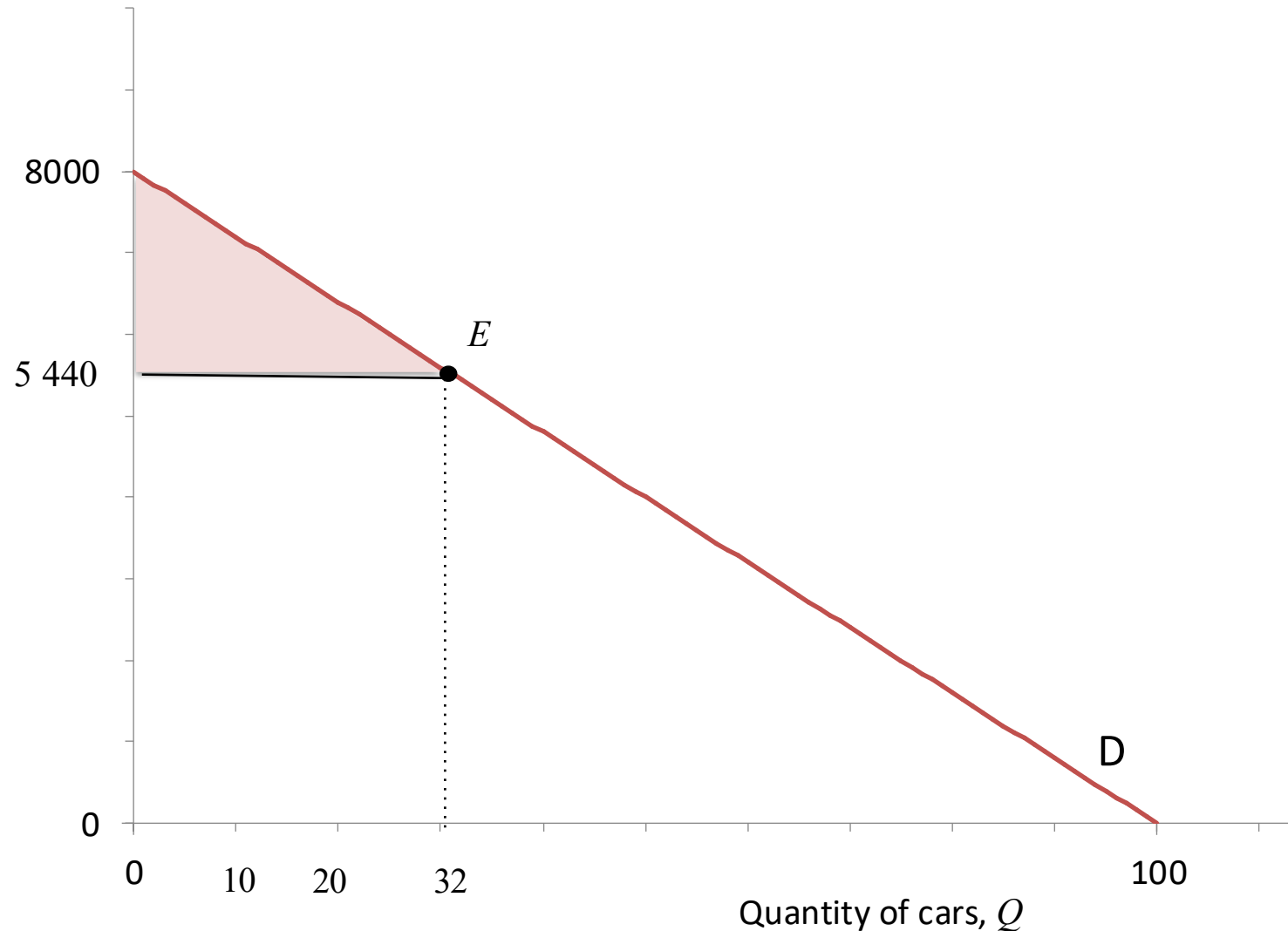
- *Consumer surplus* (CS): is obtained when a consumer pays less than what he/she is willing to pay ($WTP > P$)
- Consumer surplus = $WTP - P$
 - 10th car sold generates CS of $\$7\,200 - \$5\,440 = \$1\,760$
 - 15th car sold generates CS of $\$6\,800 - \$5\,440 = \$1\,360$
- Total CS is sum of all the CS generated by all the cars sold



Consumer surplus

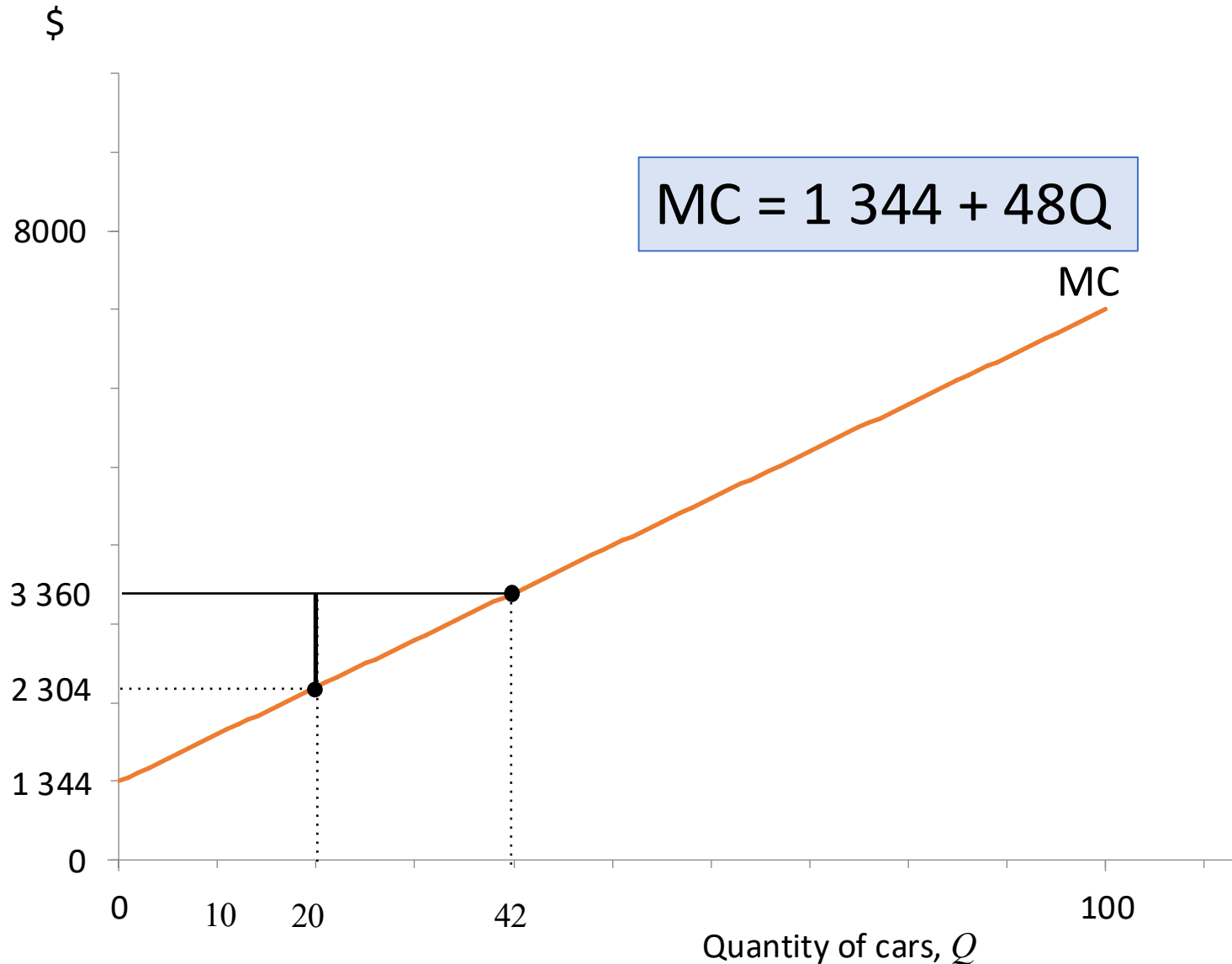
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- *Consumer surplus*: Area between demand curve and price
- Thus, the shaded area
- Consumer surplus = area of a triangle:
 $\frac{1}{2} \times 32 \times (8\,000 - 5\,440)$
 $= \$40\,960$



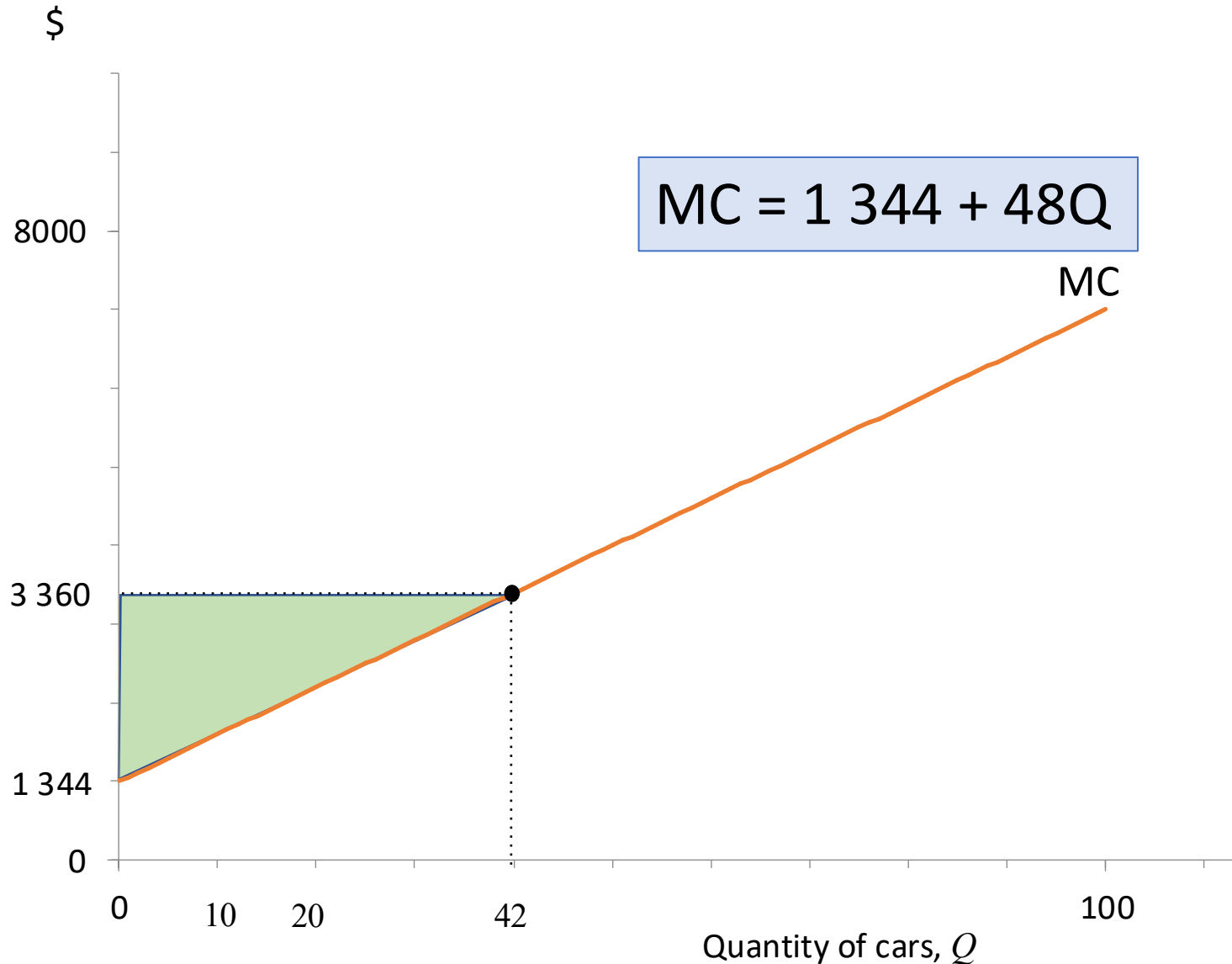
Producer surplus

- A producer is willing to sell a unit as long as the P at least covers the cost of producing that unit (MC), so $WTA = MC$
- *Producer surplus*: The amount the producer receives over and above WTA
- Producer surplus contributed by a unit $= P - MC$
- E.g. If $P = 3\,360$, producer surplus of 20th car $= 3\,360 - MC = 1\,056$
- Total PS is sum of PS generated from all units sold

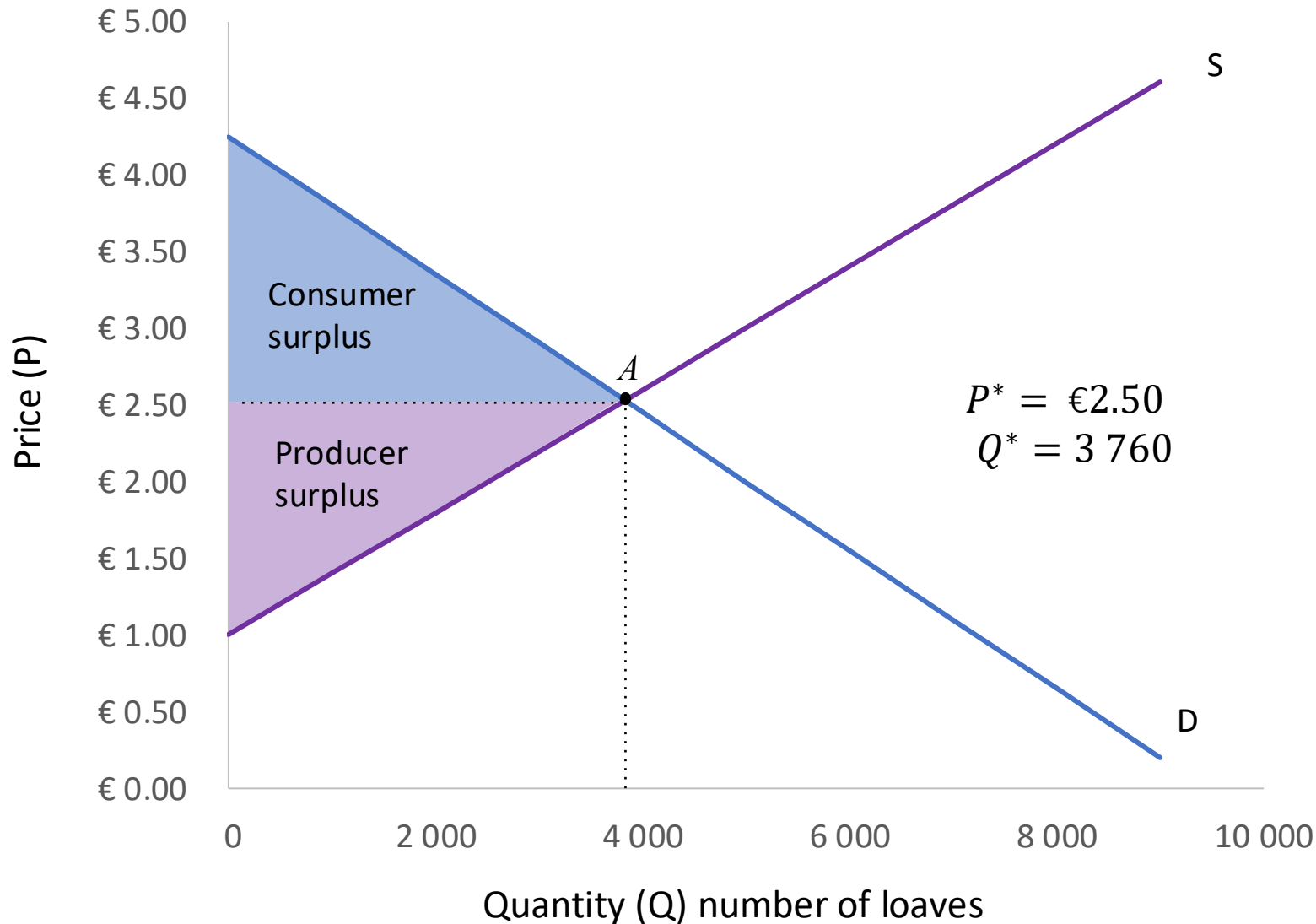


Producer surplus

- Producer surplus is the area above the MC curve and below the price
- If $P = 3\,360$ and 42 cars are sold:
$$PS = \frac{1}{2} \times 42 \times (3\,360 - 1\,344)$$
$$= \$42\,336$$

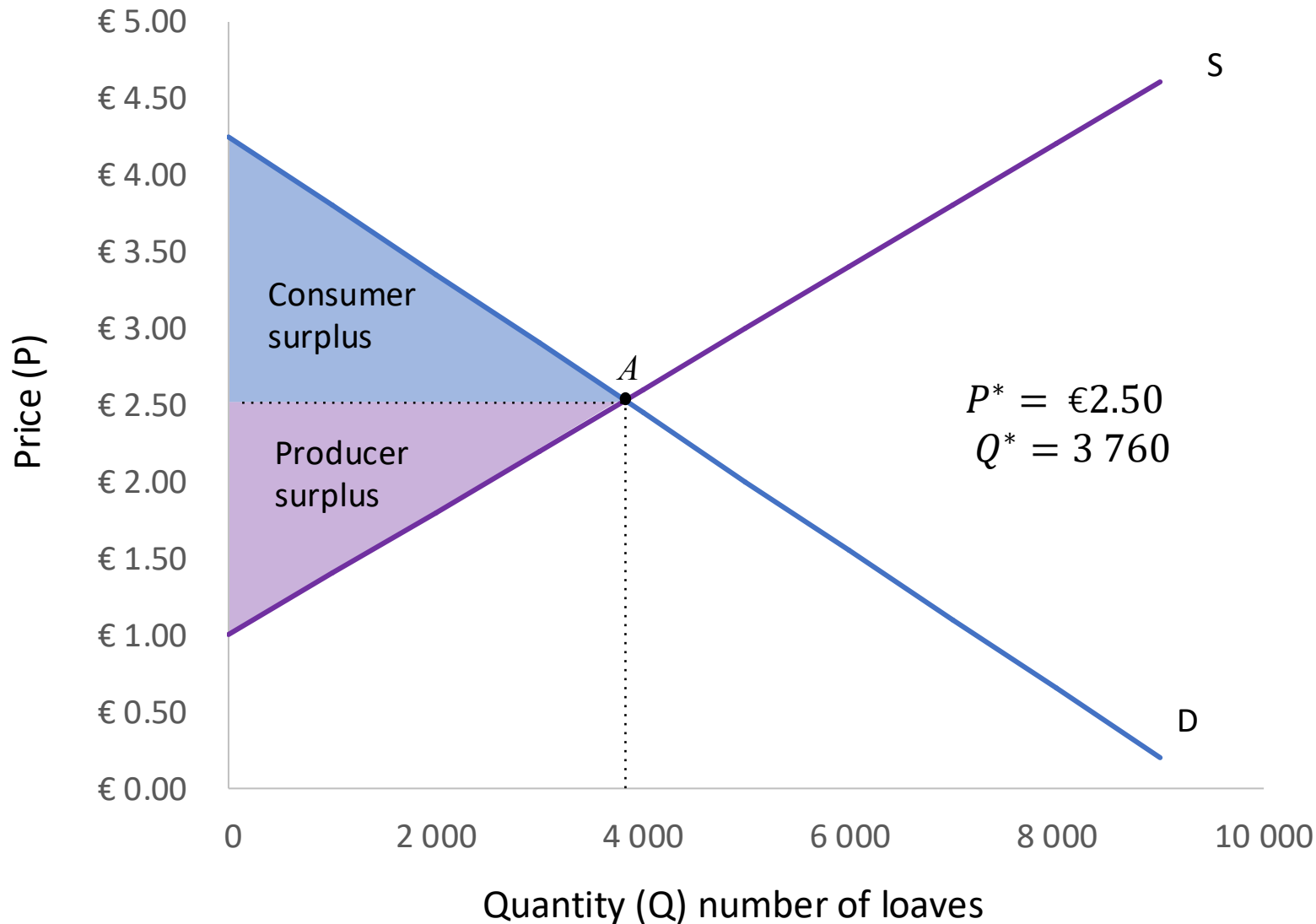


Gains from trade in competitive equilibrium



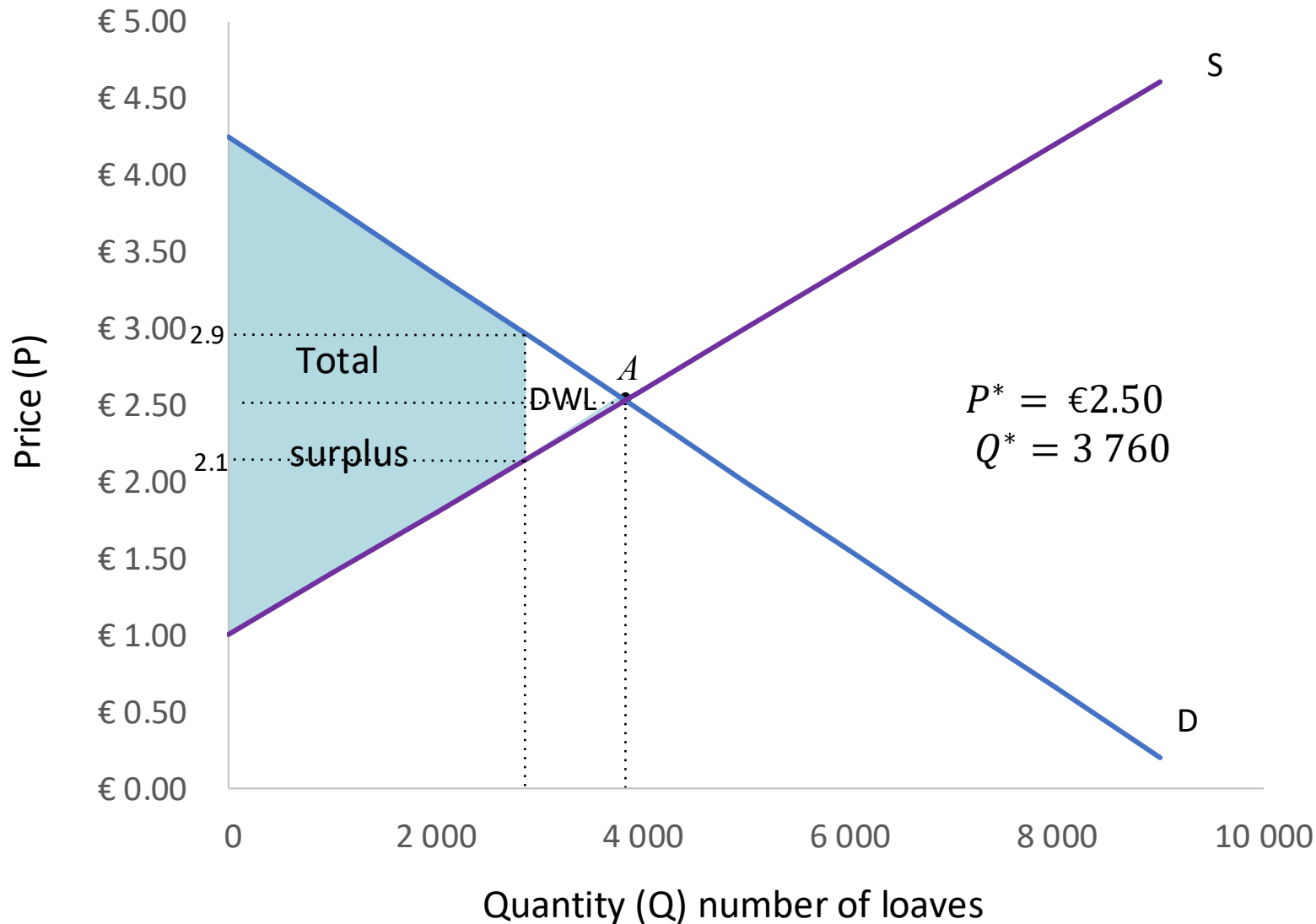
- Consider the **perfectly competitive** market for bread
- A *competitive equilibrium* allocation of bread has the property that *total surplus is maximized*
- Total surplus (TS) = CS + PS
 - Consumer surplus:
 - Buyer's WTP > price
 - $CS = (\text{€}4.25 - \text{€}2.5) \times 3\,760 \times \frac{1}{2} = \text{€}3\,290$
 - Producer surplus:
 - Seller's MC < price
 - $PS = (\text{€}2.5 - \text{€}1) \times 3\,760 \times \frac{1}{2} = \text{€}2\,820$
- TS = €3 290 + €2 820 = €6 110

Gains from trade



- When firms are price takers, all gains from trade are exhausted – total surplus cannot be increased
- The **efficient quantity** maximizes total surplus. At this quantity it is not possible to make someone better off without making someone else worse off.

Gains from trade

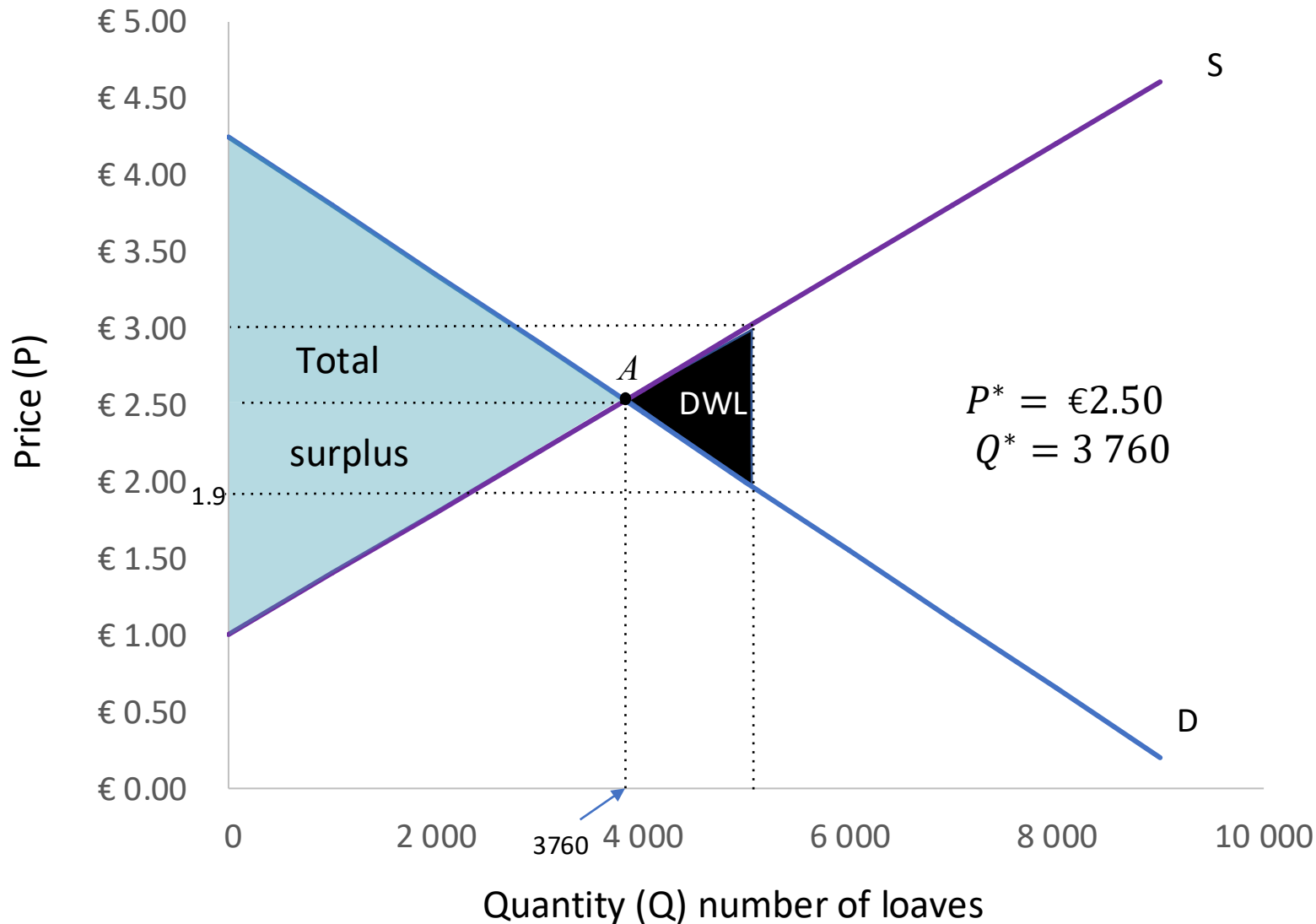


- Suppose only 2 760 loaves were exchanged (underproduction)
 - These loaves offer more benefit to consumers than what sellers are willing to accept
- For 1 000 loaves, $WTP > WTA$
- There are *unexploited gains from trade*
- The lost value from unexploited gains is called **deadweight loss (DWL)**
- $DWL =$

$$(\text{€}2.90 - \text{€}2.10) \times 1\,000 \times \frac{1}{2}$$

$$= \text{€}400$$
- This is a measure (in money terms) of inefficiency due to the quantity being different from the efficient quantity

Gains from trade

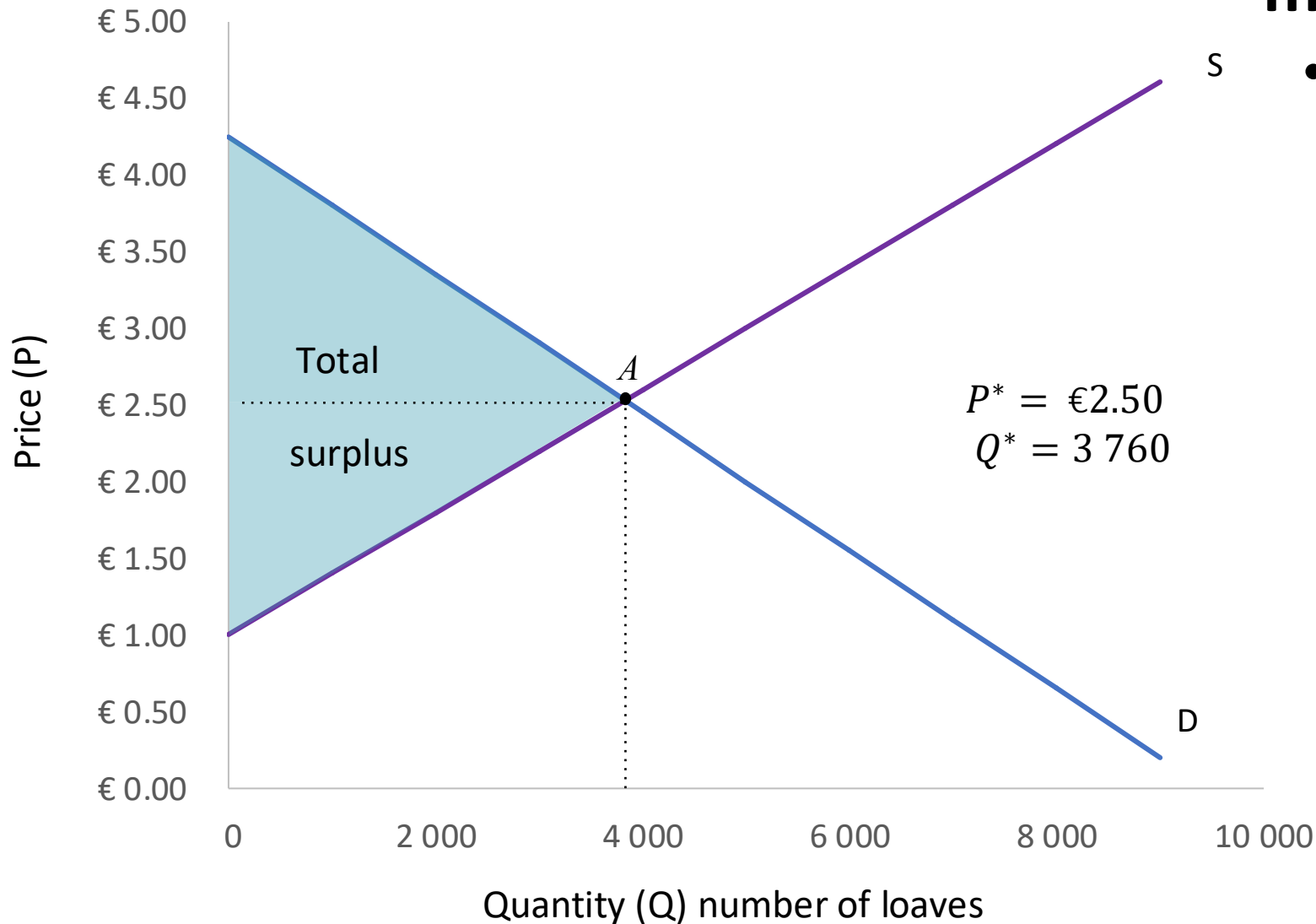


- Suppose 5 000 loaves were exchanged (overproduction)
 - These loaves cost more to produce than what consumers are willing to pay
 - For 1 240 loaves, $WTA > WTP$.
 - *Gains from trade are negative*
 - *Deadweight loss (DWL)*

$$= (\text{€}3.00 - \text{€}1.90) \times 1\,240 \times \frac{1}{2}$$

$$= \text{€}682$$

Gains from trade



Competitive markets maximize the total surplus:

- Price-taking firms sell at a price equal to MC.
- Equilibrium quantity is equal to the efficient quantity.
- The incentives lead buyers and sellers to arrive at the efficient quantity through their own individual rational choices, without any outside interference – remember Adam Smith’s “invisible hand”.
- Society allocates the efficient amount of scarce resources to the production of this particular good.

Are competitive markets a good institution?

➤ The competitive equilibrium is efficient because:

- The participants are price-takers without any market power, so that $P = MC$.
- The exchange of bread for money is governed by a complete contract between the buyer and seller.
- No effect on others besides the buyers and sellers.



Europe has competitive markets for delicious bread.

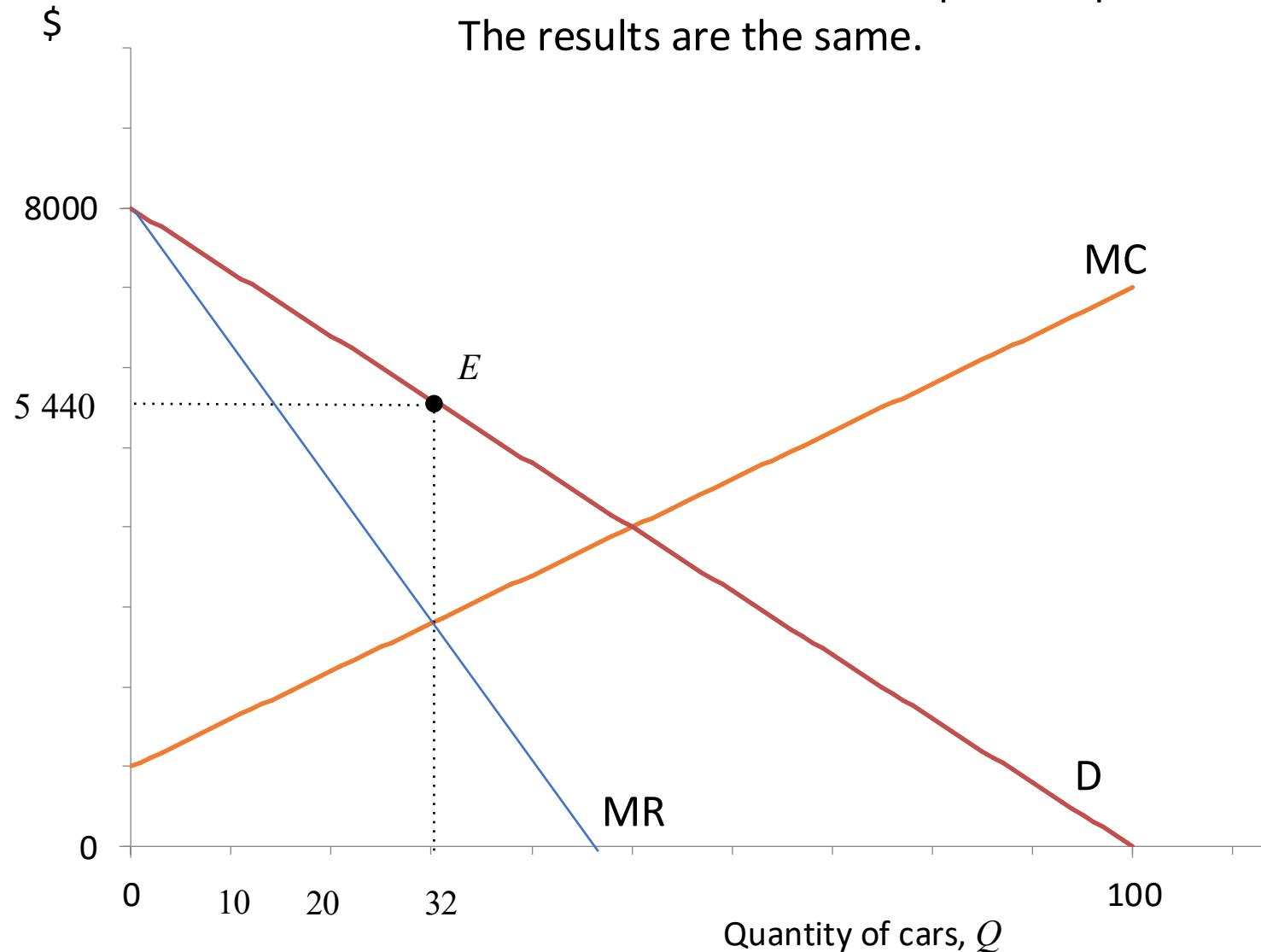
Efficiency and
equity - what
about price
setting firms?



7.7 Gains from trade

- We return to the Beautiful Cars example
- Demand curve (D) is willingness to pay (WTP)
- MC is how much it costs (ΔTC) to make an additional car ($Q+1$)
- A firm can gain from trade if it can produce a car for less than what a consumer is willing to pay
- Recall that the optimum point for the firm is where $MR = MC$
- Sell 32 cars at price of \$5 440 each

Note: the graphs in CORE uses isoprofit curves to determine the optimum point. The results are the same.

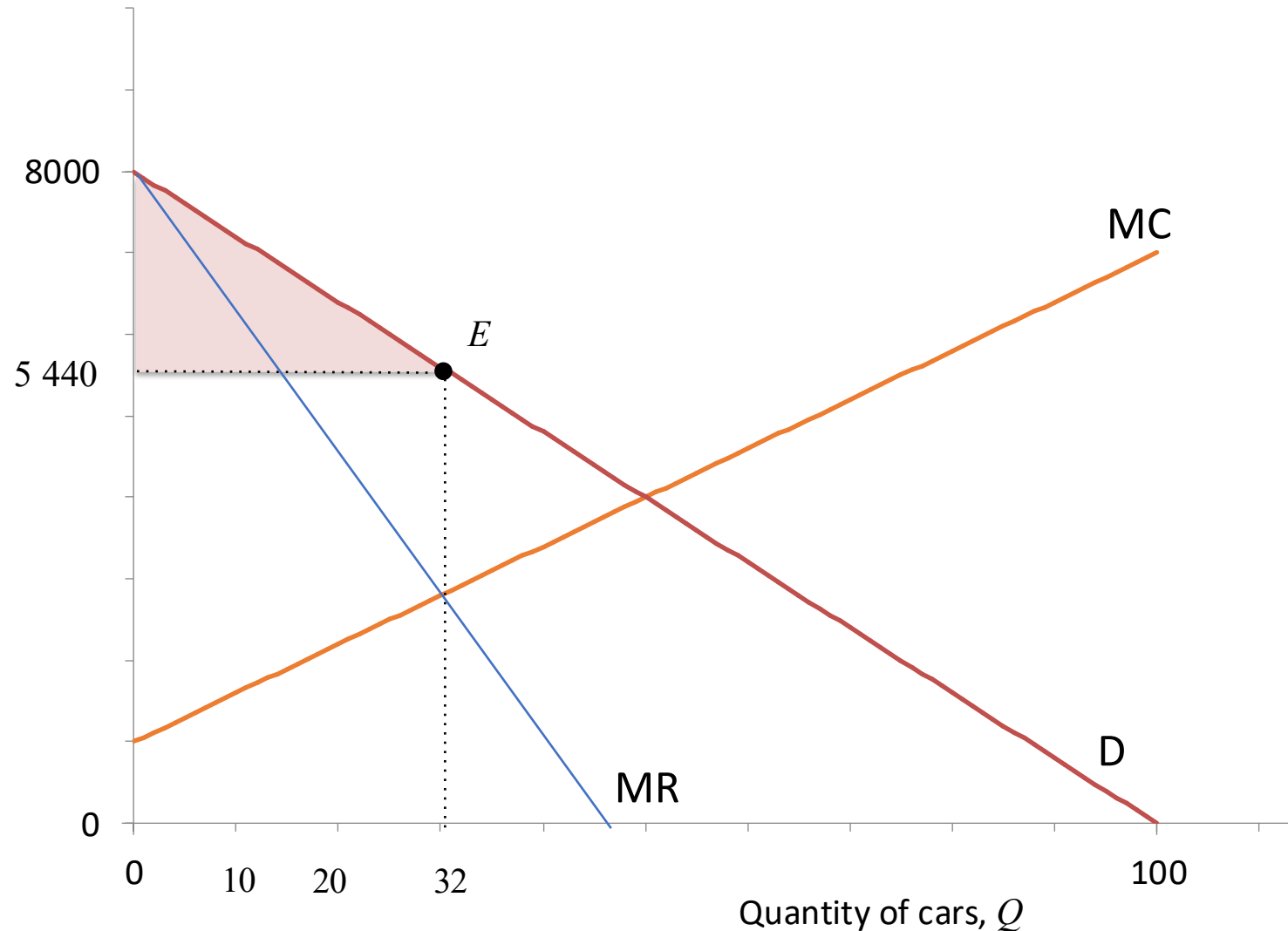


7.7 Gains from trade

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- *Total consumer surplus*: Area between demand curve and price
- Thus, the shaded area
- Total consumer surplus = area of a triangle:

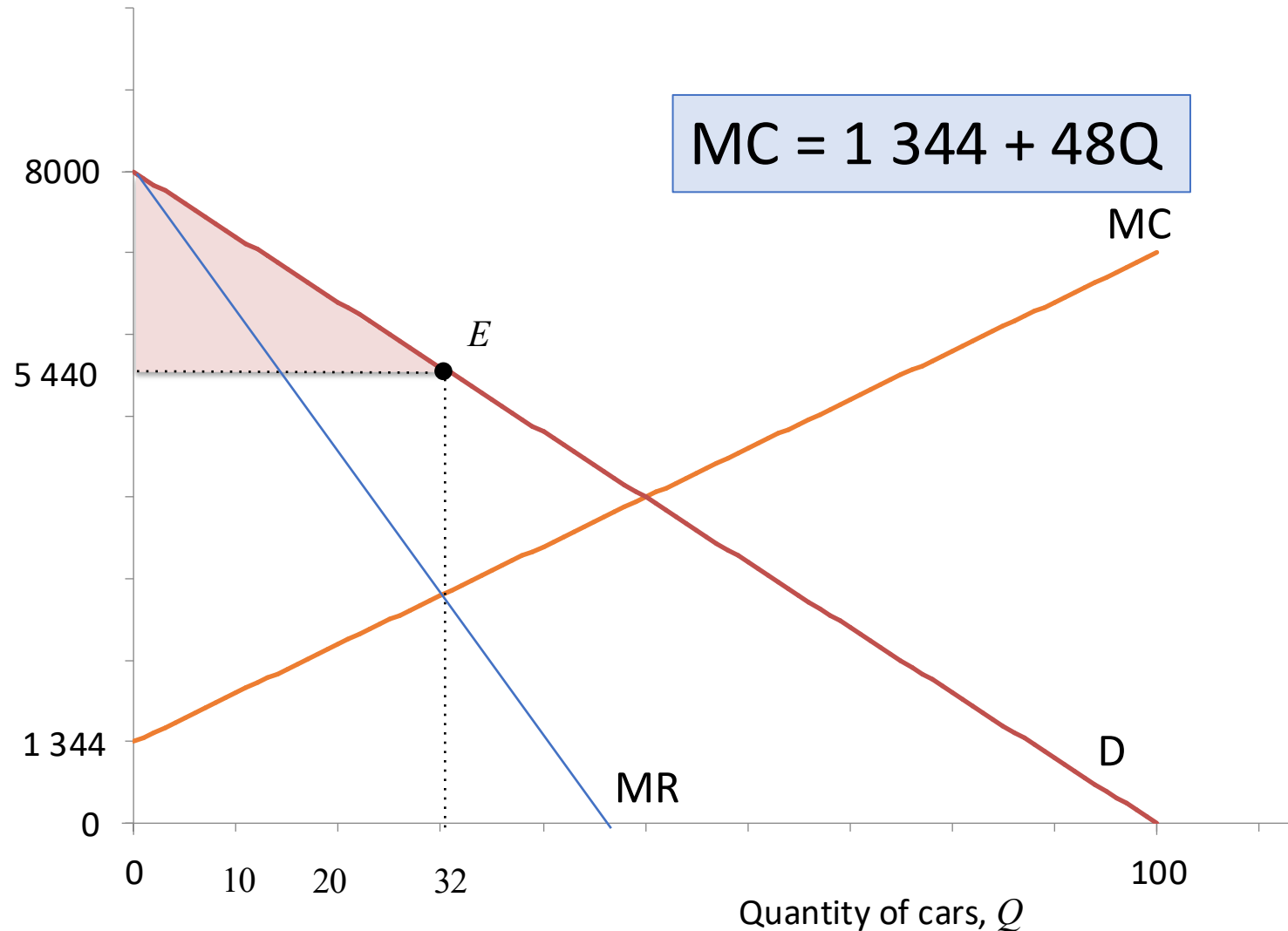
$$\begin{aligned} & (\frac{1}{2} \times 32) \times (8\,000 - 5\,440) \\ & = \$40\,960 \end{aligned}$$



7.7 Gains from trade

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- *Producer surplus*: The amount the producer receives over and above the price he is willing to accept for it (WTA)
- A producer can gain from selling another unit as long as the P at least covers the cost of producing that unit (MC)
- Producer surplus contributed by a unit = $P - MC$



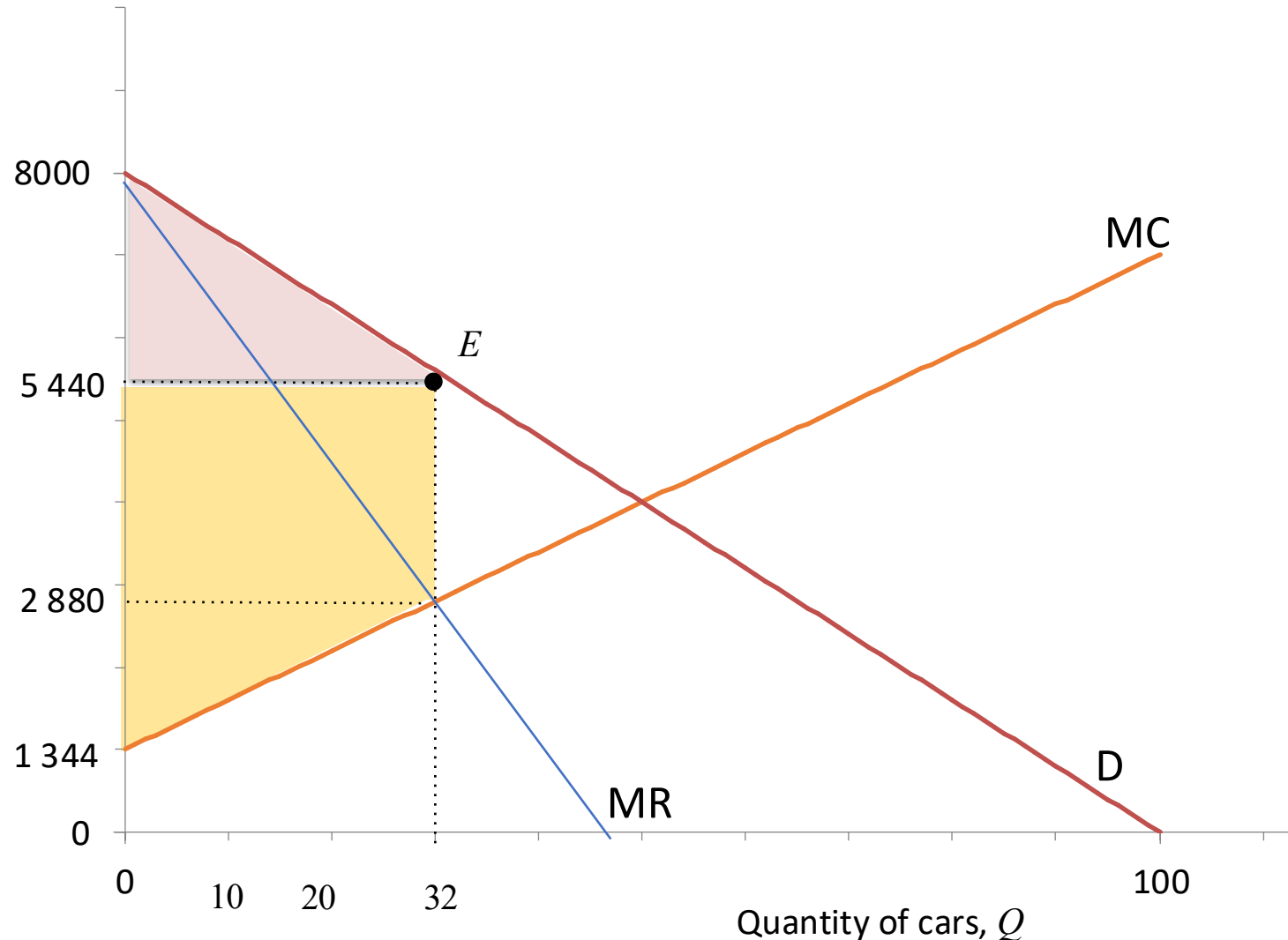
7.7 Gains from trade

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- *Producer surplus*: Area between price and MC curve
- Thus, the **yellow** shaded area of the trapezium

$$\frac{1}{2}[(5\,440 - 2\,880) + (5\,440 - 1\,344)] \times 32$$
$$= \$106\,496$$

[Area trapezium = $\frac{1}{2}(\text{sum of sides}) \times$
distance between them]



7.7 Gains from trade

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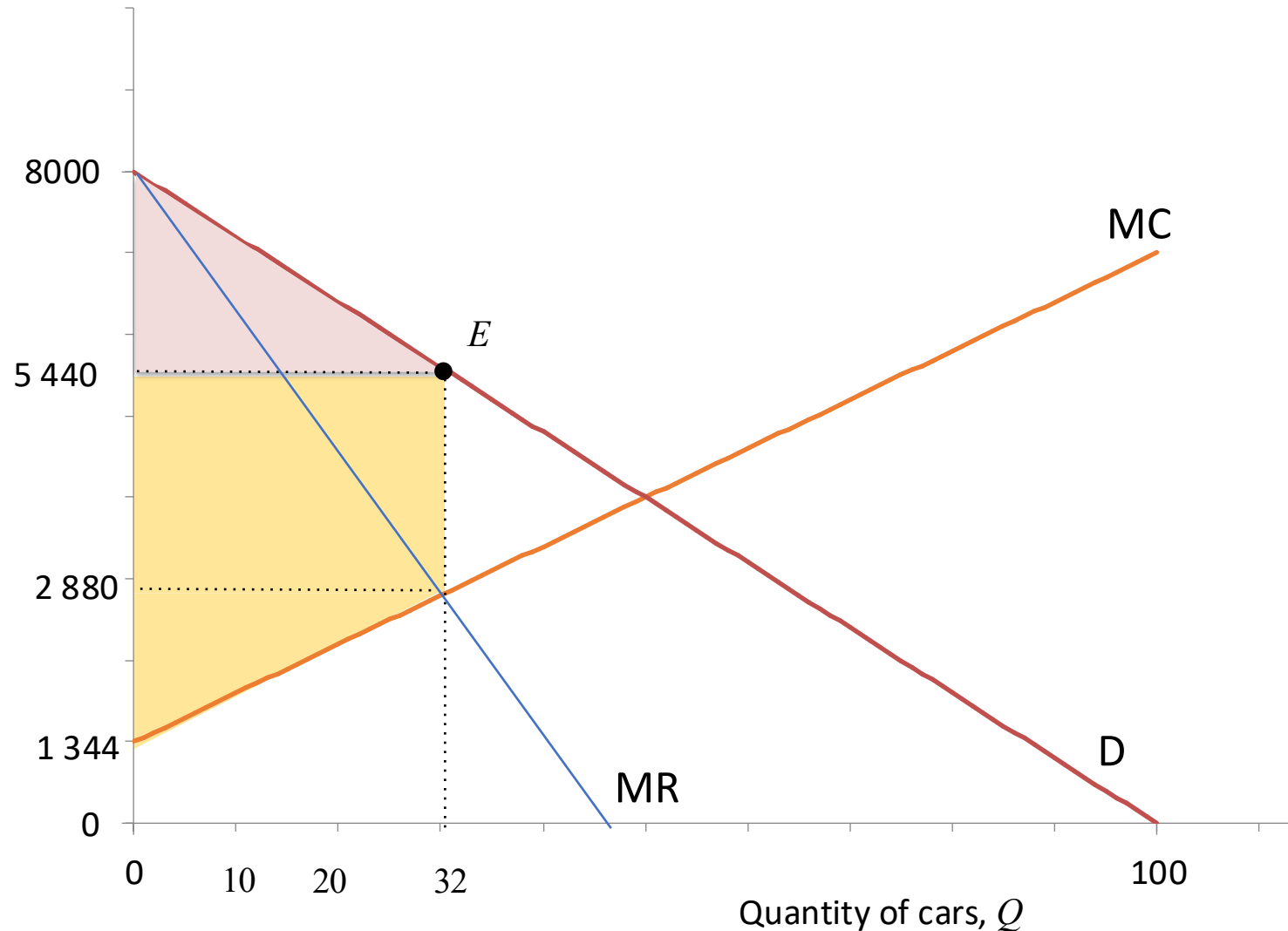
Total surplus =

consumer surplus + producer surplus

$$\$40\,960 + \$106\,496 = \$147\,456$$

- The firm is getting a larger surplus share. Why?

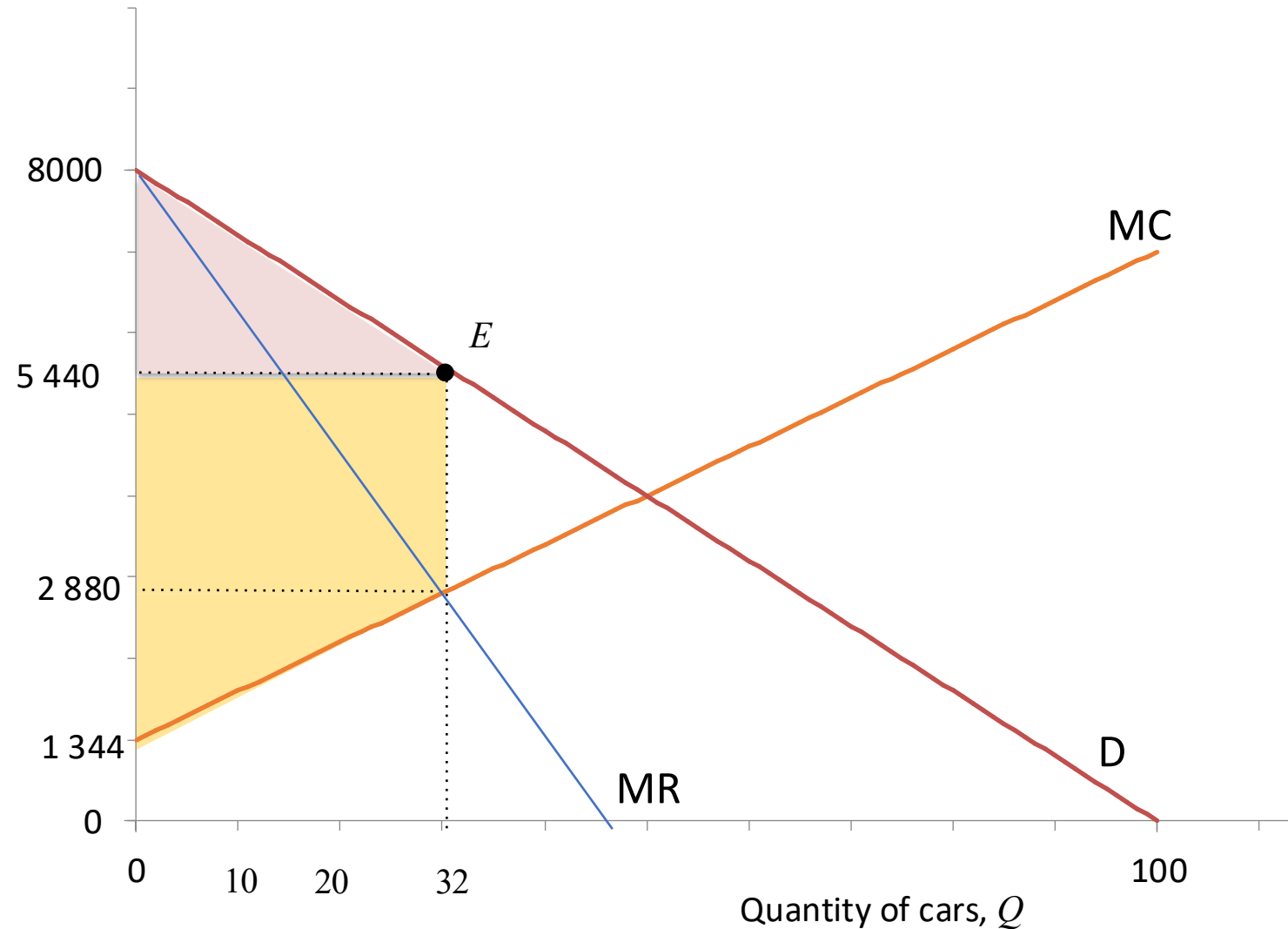
(Is this fair?)



7.7 Gains from trade

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- Does a price of \$5 440 maximize the surplus?

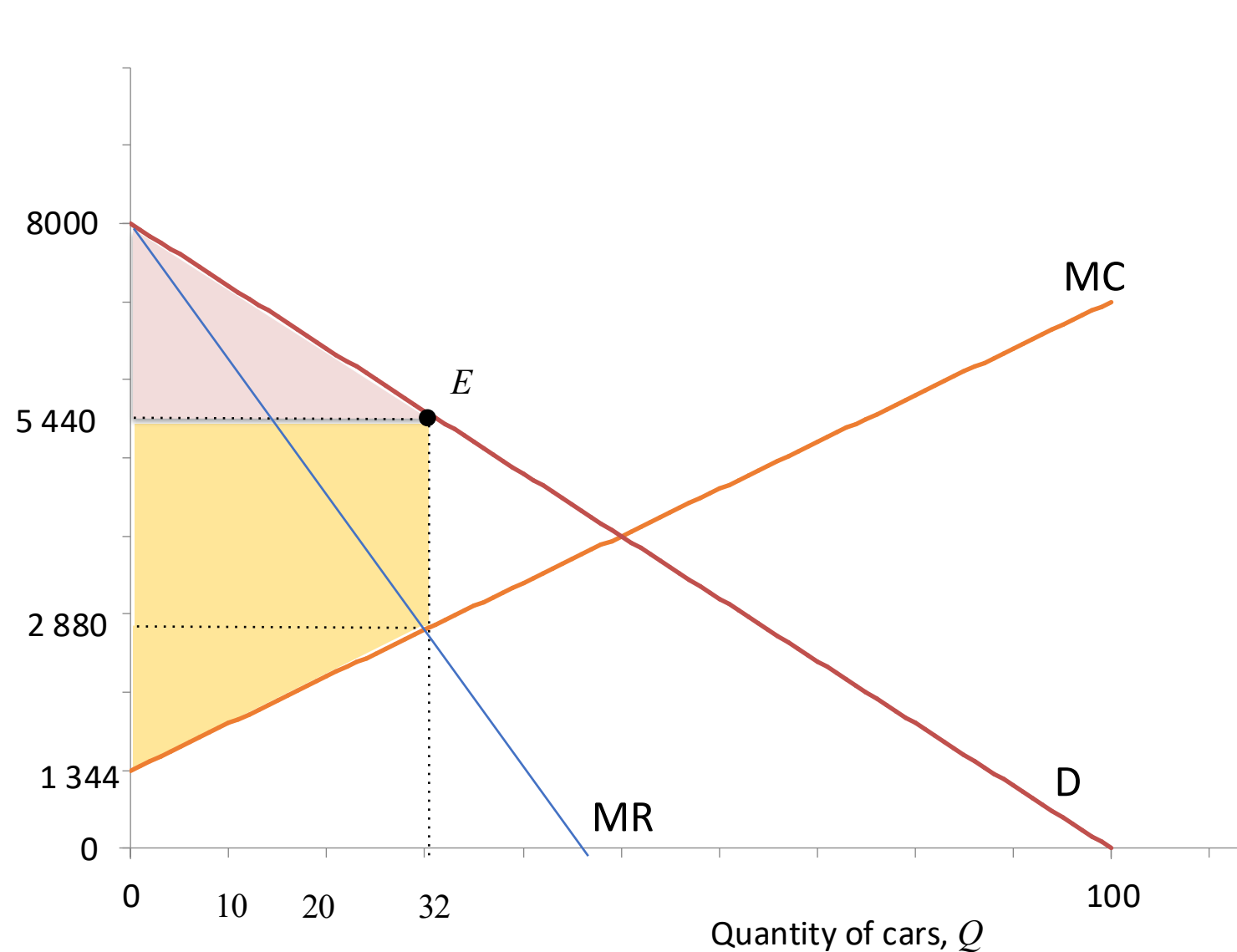


Gains from trade

- *Pareto efficiency*: no alternative allocation possible in which at least one person will be made better off, and nobody worse off.
- Is point **E** Pareto efficient?
- In the example, some consumers will be willing to pay more than it would cost the firm to produce it.
- If the firm produced another car and sold it to a 33rd customer for a price lower than \$5 440 but higher than MC, there would be a **Pareto improvement**; both the firm and the consumer would be better off.
- So point **E** is not a Pareto efficient point.

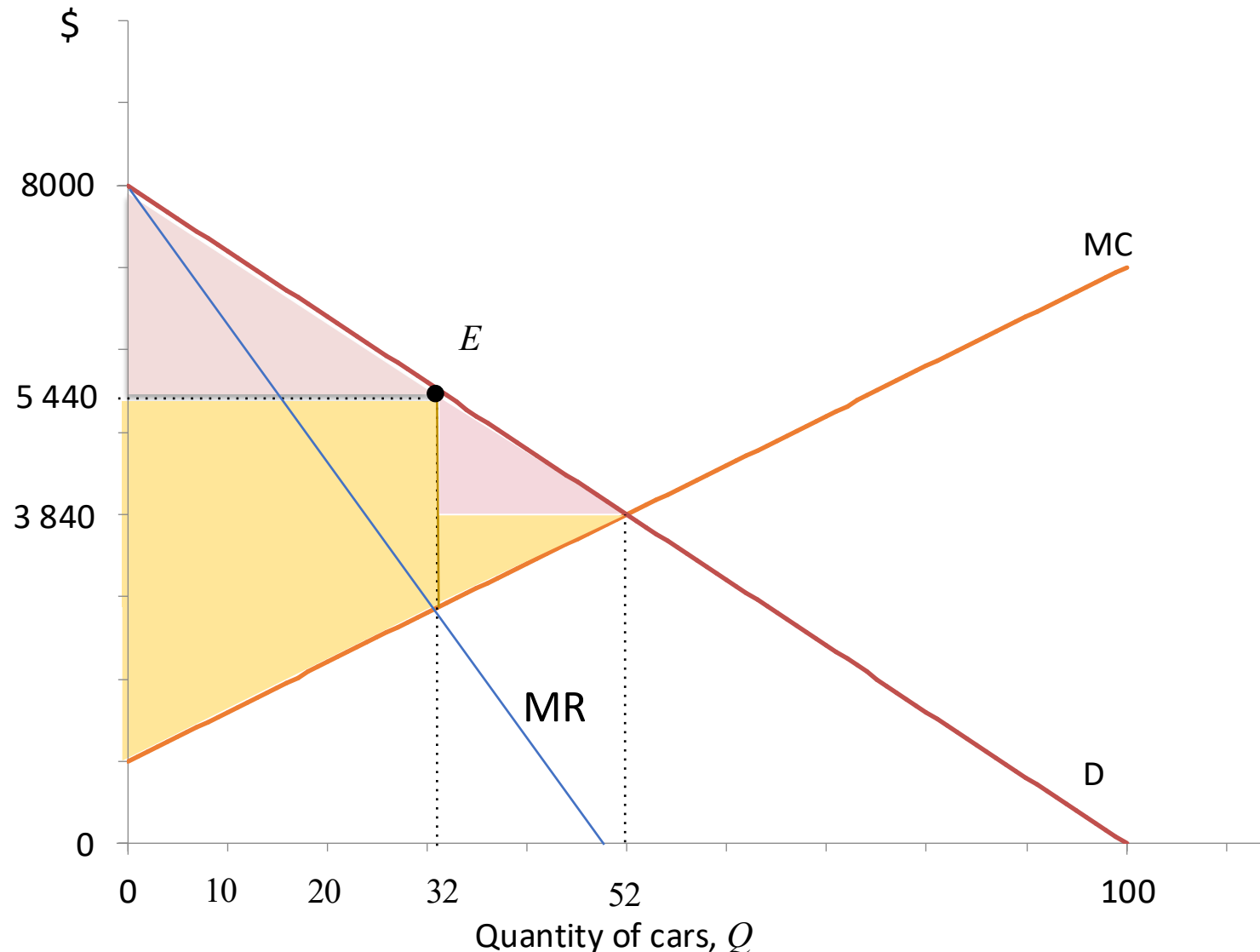
7.7 Gains from trade

- What if another car is sold?



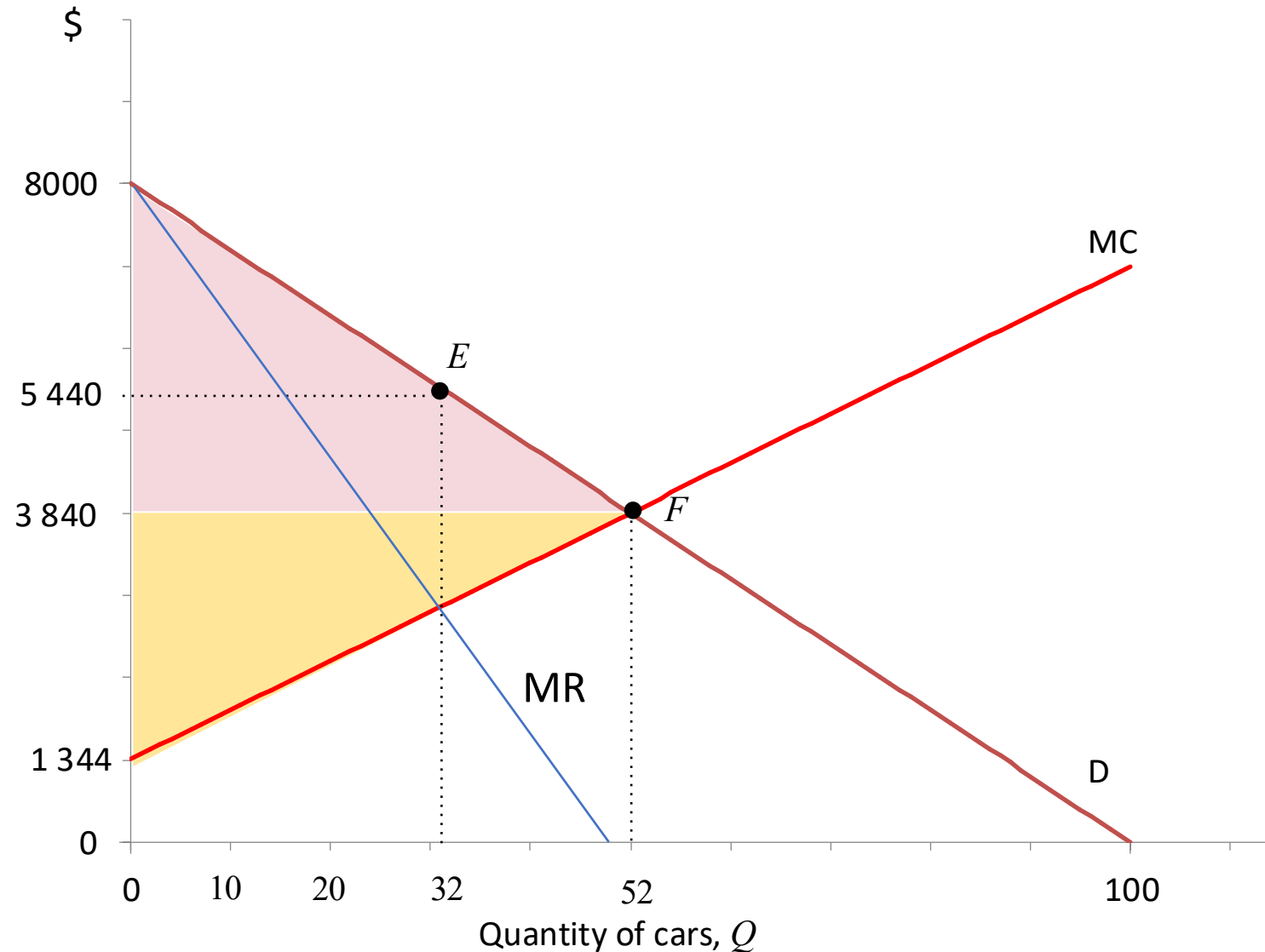
Gains from trade

- Suppose the firm sold cars to another 20 consumers for \$3 840 each (without affecting the price of the first 32 cars – the firm applies price discrimination)
- 20 consumers are better off
 - Consumer surplus increases
- The firm is better off
 - Producer surplus increases
- Gains from trade are *larger*
- Why not simply charge everyone \$3 840?



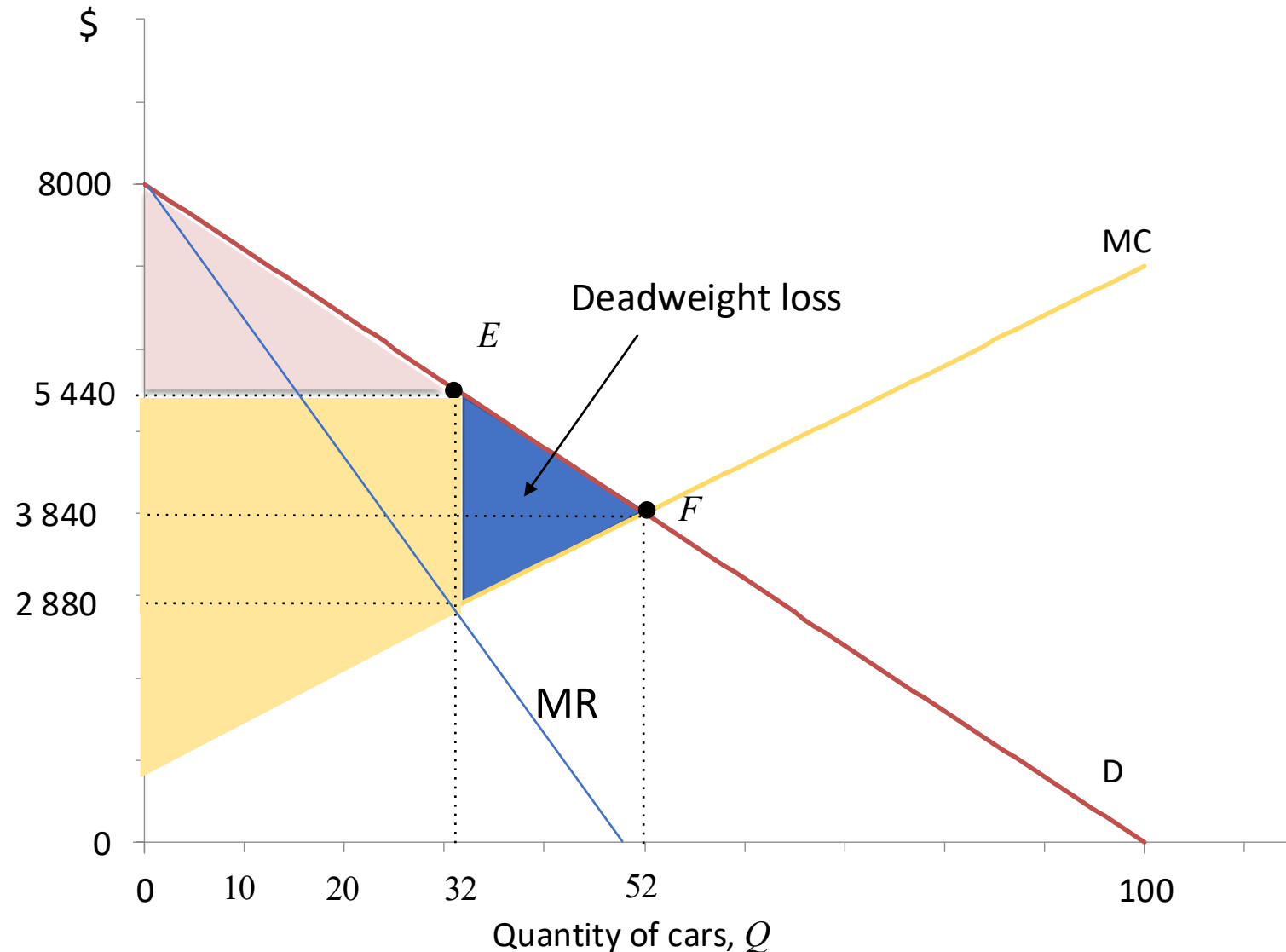
Gains from trade

- Setting the price at \$3 840 would maximize the total surplus...
 - ...but not the firm's profits.
 - How do we know this?
- The firm maximizes profits by asking a price of \$5 440



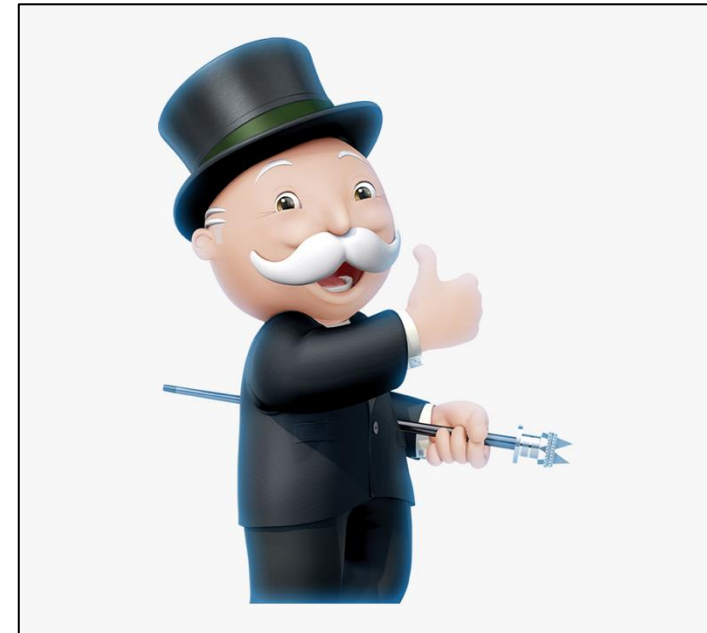
Gains from trade

- At price of \$5 440 the additional gains from trade will not be realized
- This creates a deadweight loss
 - The loss in total surplus relative to the Pareto-efficient allocation
- *Deadweight loss (DWL)* =
 $\frac{1}{2}(\$5\,440 - \$2\,880)$
 $\times (52 - 32) = \$25\,600$



Gains from trade

- Beautiful cars sell a differentiated product
- That means the firm has market power since it is the only firm producing that specific variety of the product
- Firms selling differentiated goods use their market power to set price above MC
- Monopolies (one seller of a good) do the same
- This keeps **price high** by producing a **quantity** that is **too low** relative to the Pareto-efficient allocation
- These firms' profit-maximizing incentives do **not** lead to efficient outcomes
- This is a form of **market failure**



Exercise

Cathy offers private math lessons (differentiated service). The demand for her lessons are $Q_d = 200 - 2P$. Her opportunity cost of offering lessons are given by $MC = 40 + Q$.

1. What price and quantity does she choose?
2. What is the deadweight loss resulting from her choice?

Pareto efficiency and fairness

Pareto Efficiency

- At the competitive equilibrium allocation in the bread market, it is not possible to make any of the consumers or firms better off without making at least one of them worse off
- The equilibrium allocation in the competitive market is pareto efficient

Fairness

- Recall unit 5, two criteria for assessing an allocation namely *efficiency* and *fairness*
- Although the allocation is Pareto efficient, it does not necessarily mean that it is *fair* (*i.e. distribution of surplus between rich and poor, we may want to take individual standard of living into account*)

Pareto efficiency and fairness

Pareto efficiency of a competitive equilibrium allocation is often interpreted as a powerful argument in favour of markets as a means of allocating resources.

But we need to be careful not to exaggerate the value of this theoretical result in real world markets:

- Real-world allocations may not be Pareto efficient: We might not have taken everything into account
- There are other important considerations such as fairness
- Price takers are hard to find in real life

Figure 8.20: Price-setting and price-taking firms

PRICE-SETTING FIRM OR MONOPOLY	FIRM IN A PERFECTLY COMPETITIVE MARKET
Sets price and quantity to maximise profits ("price-maker")	Takes market determined price as given and chooses quantity to maximise profits ("price-taker")
Chooses an output level at which marginal cost is less than price	Chooses an output level at which marginal cost equals price
Deadweight losses (Pareto inefficient)	No deadweight losses for consumers and firms (can be Pareto efficient if no-one else in the economy is affected)
Owners receive economic rents (profits greater than normal profits)	If the owners receive any economic rents, the rents are likely to disappear as more firms enter the market

Figure 8.20: Price-setting and price-taking firms

PRICE-SETTING FIRM OR MONOPOLY	FIRM IN A PERFECTLY COMPETITIVE MARKET
Firms advertise their unique product	Little advertising: it costs the firm, but benefits all firms (it's a public good)
Firms may spend money to influence elections, legislation and regulation	Little expenditure by individual firms on this (same as advertising)
Firms invest in research and innovation; seek to prevent copying	Little incentive for innovation; others would copy (unless the firm can succeed in differentiating its product and escaping from the competitive market)

Homework

Consider a competitive market for used copies of Greg Mankiw's textbook *Principles of Economics*. The daily supply of books is given by the equation $Q_s = P - 10$, while the daily demand for books is given the equation $Q_d = 150 - P$.

- a. Find sellers willingness to accept (or the inverse supply curve) by solving for P as a function of Q_s . What is the lowest price that anyone will accept for their book? Draw the supply curve in a figure.
- b. Find buyers willingness to pay (or the inverse demand curve) by solving for P as a function of Q_d . What is the highest price anyone will pay for a book? Add the demand curve to your figure.
- c. Find the price and quantity of textbooks in competitive equilibrium. Label both in your figure.
- d. How does competition lead buyers and sellers to converge on the equilibrium price and quantity if the initial price is higher or lower than the equilibrium price?
- e. Is the competitive equilibrium a Nash equilibrium? Why or why not?
- f. Show the consumer surplus and the producer surplus in your figure. How large is each? (*Note:* The area of a triangle is: $\text{area} = \text{base} * \text{height} * 1/2$.)
- g. Is the competitive equilibrium (Pareto) efficient? Is it fair?
- h. At the beginning of a new semester, the demand for used copies of the textbook increases to $Q_d = 350 - 2P$. Find the price, quantity, consumer surplus, and producer surplus at the new equilibrium. Is this new equilibrium efficient? Is it fair?